

Presentation 3

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them. For example,

Ben Pilnick and Zoe Lively: Mandatory Code Changes

We implemented the mandatory code changes (remove dropout from attention module, entropy regularization and gated attention). Zoe and I each ran different trials of each of the mandatory changes. I ran the code with [no dropout first](#), then with [no dropout, entropy regularization, and gated attention](#). Zoe ran gated attention [with](#) and [without](#) the new attention analysis update from Krish, as well as [with](#) and [without](#) for entropy regularization code, then ER and GA together [with](#) and [without](#) AA. This was so we can pinpoint what changes actually improved the model accuracy.

Ben Pilnick: OPTUNA_tuning

Updated Optuna_tuning files to include changes from Krish and Zoe. Krish changed the trainer function/dataset/datautilz.py files. [Optuna 2 training.py](#) includes the code with only the dropout change. [Optuna 3 training.py](#) includes Entropy regularization, gated attention, and dropout changes. Also had to add dropout for patch projector and classifier dropout. Completed 3 consecutive studies of Optuna optimization.

Isabella Toth: Best Model Visualizations & Weighted Patches Along Epithelium

For the best model, I created visualizations that reported the validation and training loss across 30 epochs, a confusion matrix of model predictions, and a deviation plot displaying model performance. Visualizations of the patches with high-attention for misclassified benign models were created. Finally, I went through Patching_Code_Attention_Annika.ipynb and worked to understand it. I altered the code slightly to make it adaptable to run with the outputs of attention_analysis.py for model runs (specifically the top_effective_patches_per_case_5.0pct.csv). Using this, I visualized the high attention patches over the slices that were misclassified by the best model.

Zoe Lively: All Benign Splits

I followed guidelines on option 2 using Krish's new code 4 and then updated it to have all [the mandatory changes](#) Ben and I added as a new folder. I then updated this code so it would test all benign splits by evaluating each benign "family" with either all in

the test or none, and getting their reported loss curve, confusion matrix, and deviation plot for each held-out family. In my presentation I give the 10 epochs early starting because the 30 epochs early starting was still running by deadline (over 12 hours), so I will update ppt with those values once done running.

Akshay Anand: Hyperparameter tuning strategy and variable vs fixed image size

I created a method that could tune the 5 most important hyperparameters while retaining computational cost as well as yielding the most accurate results for our model.

I weighed the pros and cons of fixed vs variable image sizes to determine what was the optimal strategy for our model to yield accurate results in a timely manner.

How does it help the project?

This helps the problem by allowing us to visualize which patches are getting the most attention and contributing the most to the models' decisions in a very easily interpretable way. If we can pinpoint what patches are fueling model decisions, we can determine exactly what went wrong and where when the model makes a misclassification.

The work on hyperparameter tuning and variable vs fixed image sizing can improve efficiency in the model dramatically and help provide the model with a stronger training capability.

Issues faced that haven't been resolved:

- Reduced model performance with Attention Analysis Updates
 - The model consistently underperforms compared to the original changed model with no attention analysis.
- Reduced model runtime with Attention Analysis Updates.
 - Model runtime per epoch decreased from 500-800 seconds to 60-400 seconds.
- Attention Weights in Patches Along Epithelium: All patches do not have attention weights, currently only select few.

Proposed next steps:

- Model Tuning: Tune code without the attention analysis updates for presentation
- 4. Report results.

- Visualizing Attention Weights in Patches Along Epithilium: Run code with attention weights for ALL patches, not just top ones with top/bottom attention weights
- Running/aggregating all the benign-family indices

References